Ministry of Communication and Information Technology
Digital Egypt Pioneers Initiative
(DEPI)

Coffee Shop Sales Performance Analysis Using Microsoft Power BI

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1. Introduction

This project presents a comprehensive sales performance analysis for Maven Roasters, a fictitious coffee shop operating across three locations in New York City.

The dataset consists of 149,116 transaction records covering the period from January 2023 through June 2023. Each record includes:

- Transaction date
- Transaction timestamp
- Store location
- Product-level details
- Sales amount

The purpose of this analysis is to explore sales behavior, customer purchasing patterns, product performance, and revenue distribution using Microsoft Power BI.

Through interactive dashboards and KPI tracking, this project transforms raw transactional data into actionable business insights to support strategic decision-making.

1.2 Objectives

The primary objectives of this analysis are:

- **Analyze Sales Trends Over Time:**
  - Evaluate how Maven Roasters' sales have evolved from January to June 2023.
  - Identify growth patterns, seasonal effects, and fluctuations in revenue.
- **Identify Busiest Days of the Week**
  - Determine which weekdays generate the highest traffic and sales volume.
  - Provide potential business reasoning behind these patterns.

- **Examine Peak Hours of Operation**
  - Analyze which times of day are most popular for customer purchases.
  - Compare hourly trends across all three locations to identify similarities or differences.
- **Evaluate Product Performance**
  - Identify the most and least frequently sold products.
  - Determine which products contribute the most revenue to the business.
  - Assess overall product mix efficiency.

1.3 Tools Used

- Microsoft Power BI
- Data Cleaning & Transformation
- Data Modeling
- KPI Dashboard Design
- Business Intelligence Visualization

1.4 Expected Outcome

This analysis aims to provide clear insights into:

- Revenue performance trends
- Customer behavior patterns
- Operational peak times
- Product profitability

Ultimately, this project supports data-driven decisions to improve sales performance and overall business strategy.

2. Project Planning & Management

2.1 Project Overview

This project presents a comprehensive sales performance analysis for Maven Roasters, a fictitious coffee shop operating across three locations in New York City. The dataset consists of 149,116 transaction records covering the period from January 2023 through June 2023. Each record includes transaction date, transaction timestamp, store location, product-level details, and sales amount.

The purpose of this analysis is to explore sales behavior, customer purchasing patterns, product performance, and revenue distribution using Microsoft Power BI. Through interactive dashboards and KPI tracking, this project transforms raw transactional data into actionable business insights to support strategic decision-making.

2.2 Project Scope

This project focuses on analyzing the sales performance of Maven Roasters using transactional sales data from January 2023 to June 2023.

The analysis covers the following areas:

- Overall revenue performance
- Monthly sales trends
- Sales distribution by store location
- Product category and product-level performance
- Busiest days of the week
- Peak sales hours
- Customer purchasing behavior based on transaction patterns
- Key Performance Indicators (KPIs) for business monitoring

2.3 Project Objectives

The primary objectives of this analysis are:

- Analyze sales trends over time from January to June 2023.
- Identify the busiest days of the week.
- Examine peak hours of operation.
- Evaluate product performance.
- Compare revenue performance across store locations.
- Support business decision-making through interactive Power BI dashboards.

Table 1 Project Plan

Phase	Task	Description	Deliverable
Phase 1	Data Understanding	Review dataset structure, columns, and available fields	Dataset overview
Phase 2	Data Cleaning	Clean and transform raw data using Power Query	Cleaned dataset
Phase 3	Data Modeling	Create calculated columns and DAX measures	Power BI data model
Phase 4	Dashboard Design	Build interactive dashboard pages and visuals	Power BI dashboard
Phase 5	Sales Analysis	Analyze revenue, products, locations, days, and hours	Business insights
Phase 6	Documentation	Prepare project documentation for GitHub	Final documentation files

2.4 Task Assignment & Roles

Since this is an individual project, all project responsibilities are handled by the project developer.

Table 2 Task Roles

Role	Responsibility
Data Analyst	Analyze sales trends, customer behavior, and product performance
BI Developer	Build the Power BI data model, measures, and dashboard visuals
Data Cleaner	Clean and transform the raw transactional dataset
Dashboard Designer	Design interactive and user-friendly dashboard pages
Documentation Writer	Prepare project documentation for GitHub submission

2.5 Key Performance Indicators (KPIs)

The following KPIs are used to measure and evaluate the sales performance of Maven Roasters:

Table 3 KPIs

KPI	Description
Total Revenue	Measures the total sales amount generated during the selected period
Total Transactions	Counts the number of completed transactions
Total Quantity Sold	Measures the total number of items sold
Average Order Value (AOV)	Calculates the average revenue generated per transaction
Revenue by Store Location	Compares revenue performance across the three store locations

KPI	Description
Revenue by Product Category	Identifies which product categories generate the highest revenue
Best-Selling Products	Shows the products with the highest sales quantity
Least-Selling Products	Identifies products with low sales performance
Busiest Day of Week	Determines which weekdays generate the highest sales activity
Peak Sales Hour	Identifies the busiest operating hours during the day

2.6 Tools Used

- Microsoft Power BI
- Power Query
- DAX
- Data Cleaning & Transformation
- Data Modeling
- KPI Dashboard Design
- GitHub

3. Literature Review

Business Intelligence (BI) plays an important role in helping organizations transform raw transactional data into meaningful and actionable insights. In retail and food service businesses, sales data analysis is essential for understanding business performance, revenue trends, customer purchasing patterns, and product performance.

Coffee shops generate daily transaction records that can reveal important sales patterns, such as which products are most frequently sold, which store locations generate the highest revenue, and when customers are most active during the day and week.

This project focuses on analyzing the sales performance of Maven Roasters, a fictitious coffee shop operating across three locations in New York City. The analysis is based on 149,116 transaction records covering the period from January 2023 through June 2023**.**

3.1 Business Intelligence in Sales Analysis

Business Intelligence refers to the process of collecting, cleaning, analyzing, and visualizing data to support decision-making. In sales analysis, BI tools help businesses monitor performance through key indicators such as total revenue, total transactions, total quantity sold, average order value, and product sales.

Instead of relying only on raw transaction records, BI dashboards provide a clear and interactive way to understand business performance. They allow users to filter data, compare results, identify trends, and explore performance across different dimensions such as time, location, and product category.

For **Maven Roasters**, BI analysis helps answer important business questions such as:

- How did sales change from January to June 2023?
- Which store location generated the highest revenue?
- What are the busiest days of the week?
- What are the peak sales hours?
- Which products and categories performed best?

3.2 Role of Power BI in Data Visualization

Microsoft Power BI is a business intelligence and data visualization tool used to create interactive reports and dashboards. It allows users to import data, clean and transform it using Power Query, create calculations using DAX, and build visual dashboards.

Power BI is suitable for this project because it supports:

- Data cleaning and transformation
- Interactive dashboard design
- KPI tracking
- Filtering and slicing data
- Visual comparison of sales performance
- Time-based analysis using dates and timestamps
- Product and store performance analysis

Through Power BI, the Maven Roasters dataset can be transformed from raw transaction records into clear visual insights that support business decision-making.

3.3 Importance of Sales Trend Analysis

Sales trend analysis helps businesses understand how revenue changes over time. By analyzing monthly sales from January to June 2023, Maven Roasters can identify whether sales are increasing, decreasing, or fluctuating during the analysis period.

This type of analysis is useful because it helps management:

- Monitor revenue performance
- Identify high-performing months
- Detect sales drops or unusual changes
- Understand sales behavior over time

- Support data-driven business decisions

In this project, sales trend analysis is used to evaluate the revenue performance of Maven Roasters during the first six months of 2023.

3.4 Customer Purchasing Patterns

Customer purchasing patterns can be understood by analyzing when purchases occur and what products are purchased. In a coffee shop business, time-based patterns are important because customer demand often changes during the day and across different weekdays.

For example, coffee shops may experience higher transaction activity during morning hours due to customers buying coffee before work. Other periods may show different patterns depending on customer behavior and store activity.

By analyzing transaction dates and timestamps, this project identifies:

- Busiest days of the week
- Peak hours of operation
- Transaction activity across different time periods
- Differences in sales behavior between store locations

These insights can support operational decisions such as staffing, service planning, and timing of promotions.

3.5 Product Performance Analysis

Product performance analysis helps businesses understand which products and categories contribute most to sales. In coffee shops, different products may have different levels of demand, sales quantity, and revenue contribution.

Analyzing product performance allows management to identify:

- Best-selling products
- Least-selling products
- High-revenue product categories
- Product contribution to total sales
- Customer product preferences based on transaction data

For Maven Roasters, this analysis helps determine which products are most important to the business and which products may require more attention from a sales performance perspective.

3.6 Store Location Performance

Since Maven Roasters operates across three locations in New York City, comparing store performance is an important part of the analysis. Store-level analysis helps determine whether all branches are performing similarly or whether some locations generate higher revenue and transaction volume than others.

This type of analysis can help management:

- Identify the best-performing location
- Compare revenue between branches
- Compare transaction activity across locations
- Understand location-based sales patterns
- Detect underperforming stores that may need further review

In this project, store location analysis is used to evaluate how each branch contributes to total sales performance.

3.7 Key Performance Indicators in Sales Dashboards

Key Performance Indicators, or **KPIs**, are measurable values used to evaluate business performance. In sales dashboards, KPIs provide a quick summary of the most important sales metrics.

The main KPIs used in this project include:

Table 4 KPIs in Sales

KPI	Purpose
Total Revenue	Measures the total sales amount generated
Total Transactions	Counts the total number of sales transactions
Total Quantity Sold	Measures the total number of items sold
Average Order Value	Measures the average revenue per transaction
Revenue by Store Location	Compares sales performance across branches
Revenue by Product Category	Identifies top-performing product categories
Best-Selling Products	Identifies products with the highest sales quantity
Least-Selling Products	Identifies products with lower sales quantity
Peak Sales Hour	Identifies the busiest time of day
Busiest Day of Week	Identifies the most active weekdays

These KPIs help users quickly understand the overall sales performance of Maven Roasters.

4. Requirements Gathering

The requirements gathering phase defines what the Power BI dashboard should achieve and how different users can benefit from it.

For this project, the main goal is to analyze the sales performance of Maven Roasters using transaction-level sales data from January 2023 through June 2023. The dashboard should help users understand revenue trends, customer purchasing patterns, store performance, peak hours, busiest days, and product performance.

The requirements are divided into:

- Stakeholder Analysis
- User Stories
- Use Cases

4.1 Stakeholder Analysis

Stakeholders are the individuals or groups who can benefit from the sales performance dashboard. Each stakeholder has different information needs based on their role in the business.

Table 5 Stackholder Analysis

Stakeholder	Description	Needs
Business Owner	Responsible for overall business performance and growth	Needs to monitor total revenue, sales trends, and overall business performance
Store Manager	Responsible for managing daily store operations	Needs to identify peak hours and busiest days to improve staff scheduling and service planning
Sales Manager	Responsible for tracking sales performance	Needs to compare sales by store location, month, and product category
Operations Manager	Responsible for improving operational efficiency	Needs to understand customer activity patterns by time and location
Product Manager	Responsible for product performance evaluation	Needs to identify best-selling products, low-performing products, and high-revenue categories
Data Analyst / BI User	Responsible for analyzing and presenting business data	Needs a clean, interactive, and accurate dashboard for analysis and reporting

4.2 User Stories

User stories describe the needs of the dashboard users in a simple and practical way.

- As a business owner, I want to view total revenue so that I can evaluate the overall business performance.
- As a business owner, I want to track monthly sales trends so that I can understand whether revenue is increasing or decreasing over time.
- As a store manager, I want to identify the busiest days of the week so that I can plan staffing and operations more effectively.
- As a store manager, I want to identify peak sales hours so that I can prepare for high-demand periods.
- As a sales manager, I want to compare sales performance across store locations so that I can identify the best-performing branch.
- As a product manager, I want to analyze product sales so that I can identify best-selling and least-selling products.
- As an operations manager, I want to understand sales patterns by hour and weekday so that I can improve daily operational planning.
- As a dashboard user, I want to filter the report by date, location, and product category so that I can focus on specific areas of analysis.

4.3 Use Cases

Use cases describe how users interact with the Power BI dashboard to achieve specific goals.

Table 6 Cases Analysis

Use Case	User	Description	Expected Outcome
View Sales Overview	Business Owner	The user views total revenue, total transactions, total quantity sold, and average order value	User understands overall sales performance
Analyze Monthly Sales Trends	Business Owner / Sales Manager	The user reviews sales performance from January to June 2023	User identifies sales growth, decline, or fluctuations
Compare Store Locations	Sales Manager / Operations Manager	The user compares revenue and transactions across the three store locations	User identifies the best and weakest performing locations
Analyze Busiest Days	Store Manager	The user checks which weekdays generate the highest sales activity	User improves staff scheduling and daily planning
Analyze Peak Hours	Store Manager / Operations Manager	The user reviews sales by hour of the day	User identifies high-demand operating hours
Evaluate Product Performance	Product Manager	The user reviews product sales by category, type, and product detail	User identifies best-selling and low-performing products
Filter Dashboard Data	Dashboard User	The user applies filters by date, store location, or product category	Dashboard visuals update based on selected filters

5. System Analysis & Design

This section describes the system analysis and design of the Maven Roasters Sales Performance Dashboard.

The project is designed as a Power BI business intelligence solution that transforms raw transaction-level sales data into interactive dashboards and business insights. The dashboard helps analyze revenue trends, customer purchasing patterns, store performance, product performance, busiest days, and peak sales hours.

The system is not a traditional software application with backend and frontend components. Instead, it is a data analysis and visualization system built using Microsoft Power BI, Power Query, DAX, and a structured sales dataset.

5.1 Problem Statement

Maven Roasters has a large volume of sales transaction data collected from three coffee shop locations in New York City. However, raw transaction records alone do not provide clear and direct insights into business performance.

Without proper analysis and visualization, it becomes difficult to answer important business questions such as:

- How is revenue changing over time?
- Which store location performs best?
- What are the busiest days of the week?
- What are the peak sales hours?
- Which products generate the highest sales?
- Which product categories contribute most to revenue?

Therefore, this project aims to transform the raw sales dataset into an interactive Power BI dashboard that helps users monitor KPIs, identify trends, and support data-driven decision-making.

5.2 System Scope

The system focuses on analyzing historical sales transaction data for Maven Roasters.

The dashboard includes analysis of:

- Total revenue
- Total transactions
- Total quantity sold
- Average order value
- Monthly sales trends
- Sales by weekday
- Sales by hour
- Store location performance
- Product category performance
- Product-level performance
- Interactive filtering by date, store location, and product category

5.3 Main Data Fields

Table 7 Data Field

Field	Description	Usage
Transaction Date	Date when the transaction occurred	Monthly and daily sales analysis
Transaction Time	Time when the transaction occurred	Hourly and peak time analysis
Store Location	Branch where the transaction occurred	Store performance comparison
Product Category	Main category of the product	Category-level sales analysis

Field	Description	Usage
Product Type	Type or group of product	Product type performance analysis
Product Detail	Specific product name or detail	Product-level performance analysis
Quantity	Number of units sold	Quantity sold calculation
Unit Price	Price per unit	Pricing and sales calculation
Sales Amount	Total transaction sales value	Revenue calculation

5.4 Calculated Columns

Additional calculated columns may be created in Power BI to support time-based analysis.

Table 8 Calculated Columns

Calculated Column	Purpose
Month Name	Used to analyze sales by month
Month Number	Used to sort months correctly
Day Name	Used to analyze sales by weekday
Day Number	Used to sort weekdays correctly
Hour	Used to analyze peak sales hours
Revenue	Used to calculate total sales if sales amount needs to be derived

Example calculated columns:

Revenue = 'Sales'[Quantity] * 'Sales'[Unit Price]

Hour = HOUR('Sales'[Transaction Time])

Month Name = FORMAT('Sales'[Transaction Date], "MMMM")

Month Number = MONTH('Sales'[Transaction Date])

Day Name = FORMAT('Sales'[Transaction Date], "dddd")

Day Number = WEEKDAY('Sales'[Transaction Date], 2)

5.5 DAX Measures

The dashboard uses DAX measures to calculate the main KPIs.

Table 9 DAX Measures

Measure	Formula Logic	Purpose
Total Revenue	SUM of sales amount	Measures total sales value
Total Transactions	COUNT of transaction records or transaction IDs	Measures number of sales transactions
Total Quantity Sold	SUM of quantity	Measures total units sold
Average Order Value	Total Revenue divided by Total Transactions	Measures average revenue per transaction

Total Revenue = SUM('Sales'[Sales Amount])

Total Transactions = COUNTROWS('Sales')

Total Quantity Sold = SUM('Sales'[Quantity])

Average Order Value = DIVIDE([Total Revenue], [Total Transactions])

5.6 Data Flow

Data Flow Steps:

- The raw sales dataset is imported into Power BI.
- Power Query is used to clean and transform the data.
- Data types are corrected for date, time, text, and numeric fields.
- Calculated columns are created for month, weekday, and hour analysis.

- DAX measures are created to calculate sales KPIs.
- Dashboard visuals are built using the cleaned data model.
- Users interact with slicers and visuals.
- Insights are generated from the dashboard.

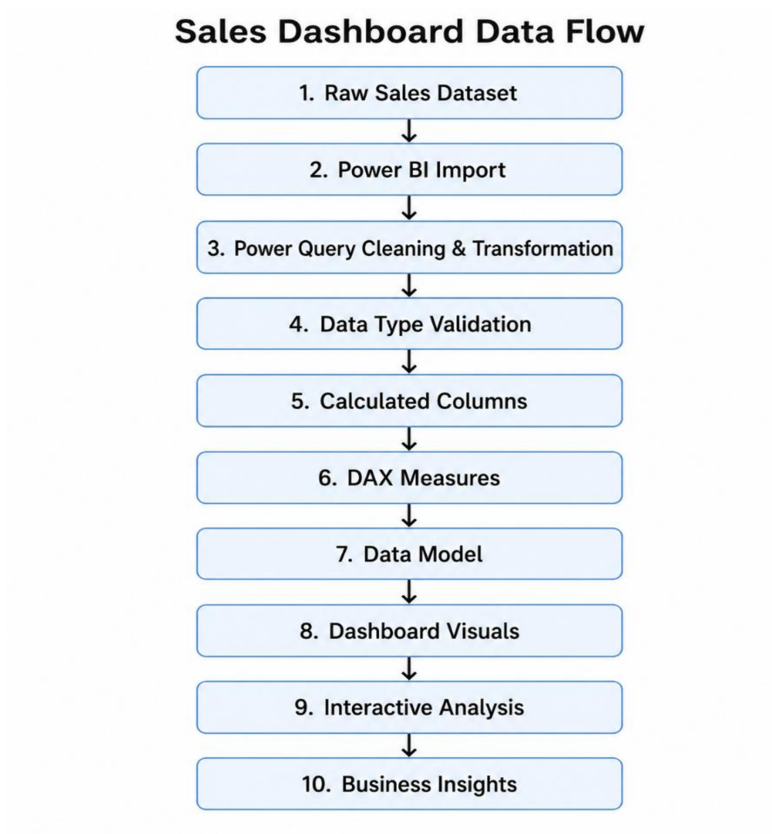

Figure 1 Dashboard Flowchart

5.7 Dashboard Pages

Table 10 Dashboard Design

Page	Purpose	Main Visuals
Executive Overview	Provides a high-level summary of sales performance	KPI cards, revenue trend, revenue by location

Page	Purpose	Main Visuals
Financial Analysis	Analyzes financial performance and key revenue indicators	Total revenue, AOV, revenue trends, revenue by product category, financial KPI cards
Product Performance	Evaluates product and category sales	Product category chart, best-selling products, least-selling products

6. Conclusion

The Maven Roasters Sales Performance Dashboard project provides a clear and interactive Power BI solution for analyzing sales performance using transaction-level data from January 2023 to June 2023.

The project starts with defining the main requirements, including stakeholder needs, user stories, use cases, functional requirements, non-functional requirements, data requirements, and dashboard requirements. These requirements help ensure that the dashboard is designed to answer key business questions related to revenue, transactions, quantity sold, average order value, store performance, product performance, busiest days, and peak sales hours.

The system analysis and design section explains how the raw sales dataset is transformed into useful business insights. The process includes importing the data into Power BI, cleaning and transforming it using Power Query, validating data types, creating calculated columns, building DAX measures, designing the data model, and presenting the results through interactive dashboard visuals.

The dashboard is organized into clear pages, including Executive Overview, Time Analysis, Store Performance, and Product Performance. Each page focuses on a specific analysis area, allowing users to move from a general business summary to more detailed insights. The dashboard also supports interactive filtering by date, store location, and product category, which allows users to explore the data more effectively.

Overall, this project demonstrates how Power BI can be used to convert raw sales data into a professional business intelligence report. The final dashboard supports better decision-making by helping stakeholders monitor performance, identify trends, compare store locations, evaluate product sales, and understand customer purchasing behavior.